Erdős 252

Irrationality of the factorial divisor-sum series

Tokengrinder

Proof outline. This is a mathematical exposition of the accompanying Lean formalization. It describes the main constructions and the contradiction; the complete rigorous proof, including all estimates and finite arithmetic, is the Lean development linked below.

For a positive integer n and $k \in \mathbb{N} = \{0, 1, 2, \ldots\}$, put

$$\sigma_k(n) = \sum_{d|n} d^k, \qquad \alpha_k = \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n!}.$$

The series converges for every k. In the formal statement the sum starts at $n = 0$; the corresponding divisor-sum term is zero.

Theorem. *For every $k \in \mathbb{N}$, the real number α_k is irrational.*

Positive exponents

Fix $k \geq 1$ and suppose, for a contradiction, that α_k is rational. Clearing its denominator by a sufficiently large factorial shows that

$$T_n = n! \sum_{m>n} \frac{\sigma_k(m)}{m!} \in \mathbb{Z} \qquad \text{for all sufficiently large } n.$$

All grids, coefficients, shifts, and moduli below are fixed once k is fixed.

1. Expand the tails. Let $\left\{ {a \atop b} \right\}$ denote a Stirling number of the second kind. Expanding the first $k+1$ factorial denominators in reciprocal powers gives

$$T_n = \sum_{h=1}^{k+1} \sum_{\ell=1}^{k+1} \left\{ {\ell-1 \atop h-1} \right\} \frac{\sigma_k(n+h)}{(n+h)^\ell} + O_k(n^{-3/2}).$$

Pairing each divisor with its complementary divisor bounds the divisor count by a constant times $\sqrt{n}$. Consequently $\sigma_k(n) = O_k(n^{k+1/2})$. Together with a geometric bound for the omitted tail, this controls both sources of expansion error.

2. Align shifted arguments. Use the finite grid $E = \{0, 1, \ldots, k+1\}^k$. One explicit choice in the formalization is

$$b = k+2, \qquad F = b^k - 1, \qquad D = F!, \qquad B = 1 + kDF,$$

$$p_e = B + D \sum_{j=0}^{k-1} e_j b^j, \qquad t_e = D \sum_{j=0}^{k-1} (j+1) e_j b^j \qquad (e \in E).$$

The integers p_e are pairwise coprime, and the shifts $r_{e,h} = hp_e - t_e$ are positive for $1 \leq h \leq k+1$. The Chinese remainder theorem supplies a progression $N = A + Qm$, $A, m \in \mathbb{N}$, with $Q = \prod_{e \in E} p_e^2$, satisfying $N \equiv t_e \pmod{p_e^2}$ for every e. For sufficiently large N, the integer $n_e = (N - t_e)/p_e$ is positive. The squared congruence gives $p_e \mid n_e$. Every retained h divides D, while $\gcd(p_e, D) = 1$, so $\gcd(p_e, n_e + h) = 1$. Multiplicativity gives

$$\sigma_k(p_e)\sigma_k(n_e + h) = \sigma_k(N + r_{e,h}) \qquad (1 \leq h \leq k+1).$$

None of the multipliers p_e is required to be prime.

3. Cancel the large terms. Assign the integer binomial weights

$$w_e = \prod_{j=0}^{k-1} (-1)^{k+1-e_j} \binom{k+1}{e_j}.$$

For a term with $1 \leq h \leq \ell \leq k$, varying coordinate e_{h-1} leaves the shift $r_{e,h}$ fixed. The remaining multiplier is a polynomial of degree at most k in that coordinate. Its order-$(k+1)$ finite difference vanishes. Terms with $h > \ell$ have zero Stirling coefficient. Thus every denominator order $\ell \leq k$ cancels exactly.

To name the surviving terms, let

$$I = E \times \{1, \ldots, k+1\}, \qquad c_{e,h} = w_e p_e^{k+1} \left\{ {k \atop h-1} \right\}, \qquad \rho_k(x) = \frac{\sigma_k(x)}{x^k} \quad (x \in \mathbb{N},\ x \geq 1).$$

The weighted tail $\sum_e w_e \sigma_k(p_e) T_{n_e}$ is an integer for all sufficiently large N in the progression. After cancellation it equals

$$\sum_{(e,h)\in I} \frac{c_{e,h}\rho_k(N + r_{e,h})}{N + r_{e,h}} + O_k(N^{-3/2}).$$

Since $\rho_k(x) = O_k(\sqrt{x})$, this integer tends to zero and hence is eventually zero. Multiplying by N, the error still tends to zero. Replacing $N/(N + r_{e,h})$ by 1 introduces only another vanishing error. It follows that

$$R(N) := \sum_{(e,h)\in I} c_{e,h}\rho_k(N + r_{e,h}) \longrightarrow 0 \quad \text{along } N = A + Qm.$$

4. Contradict the remaining limit. The shift $r_* = (k+1)B$, indexed by the zero vertex and $h = k+1$, occurs exactly once; its coefficient c_* is nonzero. For a positive step Q and positive offset a, the actual arithmetic-progression mean is

$$\lim_{M\to\infty} \frac{1}{M} \sum_{j=0}^{M-1} \rho_k(Qj + a) = \mu_Q(a) := \sum_{\substack{d\geq 1 \\ \gcd(d,Q)\mid a}} \frac{\gcd(d,Q)}{d^{k+1}} > 0.$$

This includes $k = 1$: summable domination is applied to averaged divisor counts, not to an unjustified uniform bound on the pointwise divisor sum.

Choose a fresh prime L larger than Q and every shift. Refining the progression by a residue $v \in \{0, \ldots, L-1\}$ changes its individual means by the factor

$$\mu_{QL}(Qv + a) = \mu_Q(a)\left(1 - L^{-(k+1)} + \mathbf{1}_{\{L\mid Qv+a\}} L^{-k}\right),$$

where $\mathbf{1}$ is the indicator of the stated divisibility condition. One residue makes $Qv + A$ divisible by L and therefore misses every positive shift. Another makes $Qv + A + r_*$ divisible by L and hits only the unique shift r_*. The two resulting means of R differ by

$$c_*\, \mu_Q(A + r_*)\, L^{-k} \neq 0.$$

But both means must be zero if $R(A+Qm) \to 0$, because the same limit holds on either subprogression and under Cesàro averaging. This is the desired contradiction.

Exponent zero

For $k = 0$, divisor pairing and a geometric estimate directly give $0 < T_n = O(n^{-1/2})$. These positive tails cannot be eventually integral: for large n they lie strictly between 0 and 1. Rationality would force precisely that eventual integrality, so α_0 is irrational too.

Formal verification and sources

The complete theorem is `Erdos252.erdos_252`. Its checked axiom dependencies are exactly `propext`, `Classical.choice`, and `Quot.sound`. Build, statement audits, and fresh replay using Lean's kernel have passed with Lean 4.33.1 and the pinned mathlib revision. Fresh replay uses Lean's own kernel, not a separate kernel implementation. No prime-pattern conjecture, distribution assumption, or finite computational surrogate is used.

The source repository provides reproduction instructions and exact dependency revisions. Its formal entry-points are the complete theorem, the positive-exponent proof, the zero-exponent proof, and the statement audit (definitions, summability, and reindexing). See also the original problem statement.